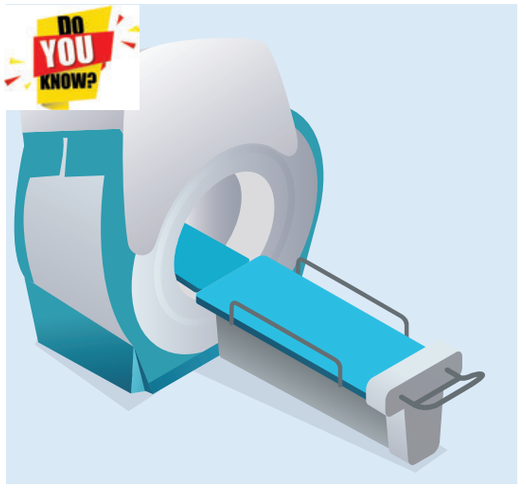



MRI is Magnetic Resonance Imaging which helps the physicians to diagonise or monitor treatment for a variety of abnormal conditions happening within the head, chest, abdomen and pelvis. It is a non invasive medical test. The patient is placed in a circular opening (actually interior of a solenoid which is made up of superconducting wire) and large current is sent through the superconducting wire to produce a strong magnetic field. So, it uses more powerful magnet, radio frequency pulses and a computer to produce pictures of organs which helps the physicians to examine various parts of the body.

(a) length of the solenoid becomes twice and fixed number of turns
L→2L (length becomes twice)
N→N (number of turns are fixed)
The magnetic field is

$$B_{2L,N} = \mu_\circ \frac{NI}{2L} = \frac{1}{2} B_{L,N}$$

(b) both the length of the solenoid and number of turns are double
L→2L (length becomes twice)
N→2N (number of turns becomes twice)
The magnetic field is

$$B_{2L,2N} = \mu_\circ \frac{2NI}{2L} = B_{L,N}$$

(c) the number of turns becomes twice but for the fixed length of the solenoid
L→L (length is fixed)
N→2N (number of turns becomes twice)
The magnetic field is

$$B_{L,2N} = \mu_\circ \frac{2NI}{L} = 2B_{L,N}$$

From the above results,

$$B_{L,2N} > B_{2L,2N} > B_{2L,N}$$

Thus, strength of the magnetic field is increased when we pack more loops into the same length for a given current.

## 3.9.5 Toroid

A solenoid is bent in such a way its ends are joined together to form a closed ring shape, is called a toroid which is shown in Figure 3.47. The magnetic field has constant magnitude inside the toroid whereas in the interior region (say, at point P) and exterior region (say, at point Q), the magnetic field is zero.

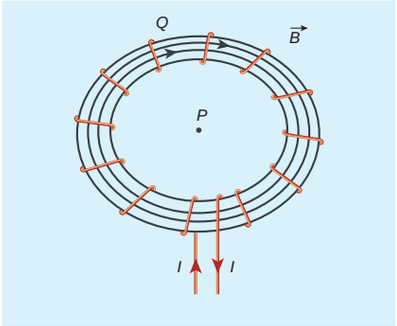



**Figure 3.47** Toroid



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